

SHUBHANKAR GUPTA

AI Engineer — LLM Applications · RAG & Evaluation · Agent Tooling (MCP) · Python · Production Systems

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github.com/shubhankar360 | linkedin.com/in/shubhankar-gupta-a73aa21a0 | shubhankar360.github.io — portfolio | solquara.com — live production platform I built and run

SUMMARY

AI engineer who ships LLM systems end to end and runs them in production. I built and operate the whole platform behind **solquara.com**, an AI video-ad and automation studio — a 40-route site, an accounts and entitlements system covered by **125 end-to-end checks**, a generative-media pipeline, and an MCP server that lets Claude drive Gemini, Veo and TTS. I publish evaluation work with the numbers: **agentcheck** showed a transcript-only grader agreeing with ground truth worse than chance on a tool-using agent, and a retrieval benchmark of mine **contradicted the hybrid-search consensus**. I profile before optimising, evaluate before claiming quality, and make every repo run from a clean checkout with no API key, with CI to prove it. Self-taught, after eight years running operations and client service for a family business.

TECHNICAL SKILLS

LLM applications: Anthropic Claude · OpenAI API · Gemini API · LLMs · tool / function calling · structured JSON output · prompt engineering · multimodal generation (image, video, TTS)
Retrieval (RAG): RAG · hybrid search (BM25 + dense) · reciprocal rank fusion · cross-encoder reranking · embeddings · citation grounding & refusal guards
Evaluation: LLM & retrieval evaluation · pass@k, Cohen's κ, McNemar & bootstrap CIs · LLM-as-judge calibration · Recall@k, MRR, nDCG · regression suites in CI
Agent tooling: Model Context Protocol (MCP) · JSON-RPC · long-running jobs (poll & resume)
Backend & data: Python · FastAPI · Pydantic · Streamlit · REST APIs · PostgreSQL (full-text, BM25, pgvector / HNSW) · Docker / Compose · SQL / SQLite / MySQL · Node.js · PHP 8 / WordPress plugins · webhooks (HMAC signing, replay protection) · web crawling & data pipelines
ML: scikit-learn · pandas · NumPy · NLP · classification & regression
Web & delivery: JavaScript (ES2022) · TypeScript · WebGL / GLSL · performance profiling · GitHub Actions CI · Git
Learning now, not yet in production: LangChain / LangGraph · LlamaIndex · Chroma / FAISS

EXPERIENCE

Founder & Lead Engineer — Solquara (solquara.com), AI video-ad & automation studio Jul 2026 – Present · Remote
Only engineer. Architecture, build, AI pipeline, testing, deployment and incident repair on a live commercial platform.

- Built the **accounts, plans and entitlements platform** as a PHP plugin: auth, four plan tiers, monthly usage limits, a one-time free sample keyed on user *and* canonical email, and billing webhooks with signature checks and replay protection. Wrote **125 end-to-end checks** — IDOR, sessions, login throttling, every subscription state — run in a local WordPress sandbox on real PHP 8.3, and **112 against production** through a key-protected, self-expiring test harness.
- Designed the **generative-media pipeline** that makes client ads: keyframe images, then image-to-video, then voiceover. Priced models before committing and found keyframe-then-animate **~4x cheaper** than the platform's built-in narrated-video workflow for the same footage. Built **aistudio-mcp** (below) so the Gemini, Veo and TTS steps can be driven from Claude.
- Shipped an **instant ad-concept writer**: a visitor names their business and trade and reads a storyboarded 10-second ad in about 20 seconds. Chose a deterministic generator over a per-visitor LLM call — zero marginal cost, and the same input always reproduces the same concept, which is what makes every result a shareable link. The concept becomes the brief for the free sample.
- Rebuilt the public site as **40 routes** (marketing pages, journal, nine browser-side tools, auth pages and a member dashboard) in vanilla ES2022 and WebGL, served ahead of WordPress by a custom plugin. Homepage: **23 stylesheets → 2, 26 scripts → 1, HTML 123 kB → 36 kB**.
- Profiled a scroll-driven UI and found backdrop-filter taking **~80% of the frame budget** (6 fps → 55 fps once replaced). Replaced device-spec sniffing, which provably excluded capable hardware, with a guard that measures real frame rate and degrades gracefully.
- Built a **B2B prospecting pipeline in Python**: OpenStreetMap extraction, a resumable website crawler, and Companies House verification that keeps only active limited companies with a role address. Output: **~920 verified prospects**. Also built regional pricing in 10 currencies from one formula shared by build and browser.
- Caught a failure every automated check missed: a duplicate animation left the **entire pricing table invisible while all HTML-level tests passed**. Added enforced rules and computed-style verification so content can never depend on a script succeeding.

Independent AI & Machine Learning Engineer — open-source, self-directed 2024 – Present · Remote

- Built and published the RAG, evaluation and agent-tooling repositories below. Every AI repo runs on a clean checkout with no API key through offline fallbacks, and CI asserts it on each push.
- Delivered end-to-end ML projects in Python and scikit-learn across classification, regression and NLP recommendation, each with reproducible notebooks and model comparisons.

Operations & Client Services — Family handicraft & handloom business, Delhi 2016 – 2024

- Ran daily operations end to end — inventory, fulfilment, pricing and after-sales — with no supervisor. Handled 20+ client conversations a day and used sales and stock data to decide what to buy.

PROJECTS — GITHUB.COM/SHUBHANKAR360

agentcheck — grading tool-using agents on what they did, not what they said Python · Anthropic SDK · evaluation

- Evaluation harness for tool-using agents: a deterministic back-office world with 7 tools and 36 tasks, graded on **end state**, a zero-tolerance **policy check** over the action log, and the facts in the answer. The environment enforces physics but not policy, so violations are caught rather than hidden.
- Measured a transcript-only grader agreeing with ground truth **worse than chance ($\kappa = -0.40$)** on an agent that skips checks, passing 16 runs that broke policy, and a 96% pass@1 agent falling to **75% pass@8** under repeated trials (unbiased estimators).
- Regression gate pairs an exact McNemar test by task with zero-tolerance policy checks. Claude adapter on the Anthropic SDK with a hand-owned tool loop, tested against a local mock of the Messages API; 22 tests, and CI regenerates the published report.

- Full-text, BM25 and pgvector retrieval fused with reciprocal rank fusion in **one SQL statement**, benchmarked on BEIR SciFact (300 claims, 5,183 abstracts): BM25 + vector hybrid reached **nDCG@10 0.712** vs 0.643 for vectors alone, while hybrid on Postgres's built-in ts_rank_cd made results **worse** (0.597).
- Found that pgvector's default ef_search silently caps LIMIT 100 at **40 rows** (Recall@100 0.925 → 0.875), and that raising it can make the planner drop the HNSW index on small tables; both documented with EXPLAIN.
- Same test suite runs on PGLite (Postgres 18 in WASM) and on a real pgvector server in CI, which caught two portability bugs the in-process run could not; Docker Compose stack verified in CI. MiniLM embeddings via onnxruntime-web, parallel and resumable.

- Zero-dependency Model Context Protocol server exposing nine tools: Gemini text and vision, image generation and editing, Omni Flash and Veo video with native audio, TTS voiceover, the Files API and live model discovery. JSON-RPC over stdio implemented from the spec.
- Long renders return a job id and resume later instead of timing out a conversation. Errors come back with a hint the model can act on (404 → re-list models, 429 → back off). Media is found by walking the response narrowest-scope first, so an echoed input image is never mistaken for the output.
- **12 end-to-end tests** spawn the real server and point it at a local mock of Google's API, asserting on the exact upstream requests. No key, no network. CI on Node 18, 20 and 22.

- Published the result that **contradicted the consensus**: fusion did not beat its best leg, and BM25 won Recall@3 while running **40x faster**. 27 tests; the benchmark itself runs in CI.
- Reproducible benchmark of four retrieval stacks — BM25, dense, hybrid RRF, hybrid + cross-encoder rerank — over a labelled query set, scored on Recall@k, MRR, nDCG and latency. BM25 and RRF implemented directly so a result is attributable to a stage.

- RAG service with recursive chunking, vector retrieval and answers that cite the exact passage. Measured that a similarity floor cannot detect unanswerable questions — an off-topic query scored **0.907 vs a relevant one's 0.905** — and replaced it with a vocabulary-coverage guard that separates them cleanly. **65 tests**, all offline.

- Pipeline over nine sources (Greenhouse, Ashby and Lever boards for 138 companies, Hacker News, five aggregators): **12,523 postings per run**, each with an eligibility verdict and an itemised score whose parts provably sum; SQLite history flags reposted roles and enforces per-company application caps.
- Evidence-constrained CV tailoring: selects and reorders facts from a profile and never writes new ones, enforced by a test; renders a two-page PDF with headless Chrome. 67 tests, standard library only.

- Grounds answers in a retrieved knowledge base, classifies sentiment with a chain-of-thought prompt, adapts tone, and uses a structured-output prompt to escalate and open a ticket. Runs in demo mode with no key; 19 tests in CI.

MACHINE LEARNING PROJECTS

- **Heart disease classification** — 13 clinical features, 303 patients; tuned Logistic Regression reached **88.5% test accuracy**, **F1 ≈ 0.87**, **recall ≈ 0.92**, beating Random Forest and KNN under 5-fold CV.
- **Wine quality prediction** — **13 models** benchmarked as parallel regression and classification problems; Random Forest led both at R² 0.52 and 93% accuracy / 0.95 ROC-AUC.
- **Movie recommender** — content-based recommender over TMDB 5000 with an NLP feature pipeline and cosine similarity.

EDUCATION

Bachelor of Arts, Tourism — Indira Gandhi National Open University (IGNOU)

2020

CBSE Class XII — Air Force Bal Bharati School, New Delhi

2014

Engineering is self-taught, and the evidence is public: every project above links to code, tests and CI.

ADDITIONAL

Languages: English (fluent, professional) · Hindi (native)

Remote: dedicated home office, reliable connectivity, async-first with written decisions. Available as an employee (via EOR) or contractor.